

MDP Planning as Policy Inference

David Tolpin,
Offtopia and Ben-Gurion University of the Negev
david.tolpin@gmail.com

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Abstract

We propose a formulation of MDP planning as Bayesian policy inference. Our formulation yields the optimal policy that maximizes the expected return, unlike other formulations of planning or control as inference, which maximize different objectives. Based on our formulation, we leverage variational sequential Monte Carlo for robust policy inference in discrete MDP domains. Further on, we suggest an interpretation of reward scale as a measure of uncertainty about the agent’s preferences, and introduce the notion of the optimal stochastic policy under preference uncertainty. We show that, under our formulation, of-the-shelf variational inference is applicable and efficient to find both the deterministic optimal policy and the stochastic optimal policy under preference uncertainty. We provide a reference implementation of MDP planning as policy inference under our formulation, and evaluate the algorithm’s robustness and efficiency on a number of domains ranging from grid worlds to challenging factored MDP domains.

1 Introduction

Casting Markov decision process (MDP) planning as Bayesian policy inference offers a two-fold benefit. First, it enables the use of general-purpose Bayesian inference algorithms for planning, potentially obviating the need for custom-designed and domain-specific solution methods. Second, uncertainty about the optimal policy becomes explicit and is represented as part of the inferred posterior, rather than being implicitly absorbed into approximation artifacts or heuristic regularization.

Despite these advantages, formulating MDP planning as Bayesian inference presents fundamental challenges arising from the special structure of the planning problem. Most Bayesian statistical models assume a clear separation between latent and observable random variables, and treat uncertainty as purely epistemic—each random variable has a fixed but unknown value. In contrast, MDP planning involves intrinsic aleatoric uncertainty due to stochastic environment dynamics and future action realizations. Moreover, identifying which

components of the MDP—states, actions, trajectories, or policies—should be treated as latent variables, and which (if any) are observed, is not straightforward.

Several Bayesian formulations of planning and control have been proposed, including approaches based on control-as-inference, maximum-entropy reinforcement learning, and active inference (Dayan and Hinton, 1997; Toussaint and Storkey, 2006; Ziebart, 2010; Levine, 2018). While these approaches recast planning as probabilistic inference, they do so by altering the optimization objective to fit standard Bayesian modeling assumptions. In particular, the expected return is augmented or replaced by entropy-regularized or evidence-based objectives, and stochasticity of the resulting policies is introduced as an intrinsic modeling preference rather than as a consequence of explicit uncertainty. As a result, the inferred policies are optimal with respect to surrogate objectives, and uncertainty over policies is not directly interpretable as posterior uncertainty about the solution of the original MDP planning problem.

In this work, we propose a formulation of MDP planning as Bayesian policy inference that preserves the standard MDP objective. The resulting posterior concentrates on policies that maximize expected return, while simultaneously providing a principled representation of uncertainty over policies. Unlike prior control-as-inference approaches, our formulation does not rely on entropy regularization as a surrogate objective, but instead derives policy stochasticity from uncertainty about the agent’s preferences.

Building on this formulation, we adapt variational sequential Monte Carlo (VSMC) (Naesseth et al., 2018; Maddison et al., 2017) to perform robust approximate inference over policies in discrete MDPs with stochastic dynamics. Furthermore, we interpret the reward scale as a parameter controlling the agent’s uncertainty about its own preferences, leading naturally to the notion of an **optimal stochastic policy under preference uncertainty**. This interpretation provides a Bayesian justification for stochastic optimal policies without departing from the expected-reward criterion.

Our contributions are as follows:

- A sound formulation of MDP planning as Bayesian policy inference that preserves the classical expected-reward optimality criterion.
- An adaptation of variational sequential Monte Carlo for Bayesian policy inference in discrete MDPs with stochastic transitions.
- An empirical study of reward scale as a measure of uncertainty over agent preferences, and its effect on the induced optimal stochastic policy.

2 Background

2.1 Markov Decision Process

Definition 1. *An episodic Markov decision process is a tuple (S, A, T, R, s_1, H) , where:*

- S is a set of states called the state space.
- A is a set of actions called the action space.
- $T(s, a) \rightarrow s' : S \times A \rightarrow S$ is a transition operator, generally stochastic, that returns the next state s' upon taking action a in state s . T implies the existence of the conditional probability $p(s' | s, a)$ of arriving at state s' after taking action a in state s .
- $R(s, a, s') : S \times A \times S \rightarrow \mathcal{R}$ is a reward function that returns the immediate reward upon taking action a in state s and arriving at state s' as the result.
- s_1 is the initial state.
- H is the number of state visits (steps) in an episode, called the horizon.

A policy $\pi(s) \rightarrow a : S \rightarrow A$ is an operator, generally stochastic, that returns an action a to take in state s . An equivalent representation of a policy is the probability $p_\pi(a | s)$ of selecting action a given state s . Conditional independence of the action from the history given the current state (that is, that a policy is $\pi(s_t) \rightarrow a$ rather than $\pi(s_1, a_1, \dots, s_t) \rightarrow a$) is an ubiquitous implicit assumption.

In an *episode*, the *agent* visits H states, starting with the initial state s_1 , by taking in each state but the last the action returned by the policy. A common objective in an episodic MDP is to find a policy π_* that maximizes the expected return (total reward), though other objectives exist. For policy inference, the MDP is *modelled by a simulator* that defines a generative process over trajectories by sequentially sampling

$$s_{t+1} \leftarrow \text{STEP}(s_t, a_t), \quad r_t \leftarrow \text{REWARD}(s_t, a_t, s_{t+1}) \quad (1)$$

initialized at s_1 . Policy inference algorithms interact with the MDP model exclusively through this generative mechanism.

2.2 Variational Sequential Monte Carlo

Variational sequential Monte Carlo (Naesseth et al., 2018) is an algorithm for variational inference in models amenable to inference with sequential Monte Carlo (SMC) (Gordon et al., 1993), in particular, in state space models. In standard SMC, the posterior distribution of latent variables is approximated by N weighted particles iteratively drawn from a proposal distribution. A byproduct of SMC estimation of the posterior is an estimate of the marginal likelihood of observations. Variational SMC (VSMC) views the parametrized proposal distribution as a variational family and maximizes a ‘surrogate ELBO’—the log marginal likelihood estimate computed by SMC.

Formally, let

$$p(x_{1:H}, y_{1:H}) = p(x_1) \prod_{t=1}^{H-1} p(x_{t+1} | x_t) \prod_{t=1}^H p(y_t | x_t) \quad (2)$$

be a state space model specified by the joint probability distribution, where $x_{1:H}$ are the latent variables, $y_{1:H}$ are the observations, and H is the number of steps through the state space (the horizon). SMC approximates the *evidence*, or marginal likelihood, $Z = p(y_{1:H})$ by a set of N *particles* and their weights $w_{t,i}$ at each step, as

$$\hat{Z} = \prod_{t=1}^H \frac{1}{N} \sum_{i=1}^N w_{t,i}, \quad (3)$$

where the weights are computed based on conditionals $p(x_{t+1} \mid p_s)$, $p(y_s \mid x_s)$ and proposal $q(x_{t+1} \mid x_s, \lambda)$.

$\hat{Z}$ is an unbiased estimate of Z but $\log \hat{Z}$ is a biased estimate of $\log Z$ due to the concavity of $\log$, and the bias depends on the SMC proposal. The proposal that minimizes the bias is the conditional distribution $p(x_{t+1} \mid x_t, y_{t:H})$. Thus, the VSMC-inferred proposal is an approximation of the conditional posterior distribution $p(x_{t+1} \mid x_s, y_{1:H})$ (x_{t+1} is independent of $y_{1:t-1}$ given x_t).

3 Probabilistic Model

A widespread setting for posterior Bayesian inference is a graphical model involving a latent variable x , an observable variable y , and a joint distribution $p(x, y)$. By Bayes' rule,

$$p(x \mid y) = \frac{p(x, y)}{p(y)}, \quad (4)$$

and upon observing $y = y^*$, one obtains an unnormalized posterior over x ,

$$p(x \mid y = y^*) \propto p(x, y^*). \quad (5)$$

Posterior inference methods then aim to characterize this distribution, exactly or approximately, by producing samples or expectations consistent with the corresponding normalized distribution. Although most practical algorithms do not evaluate the normalizing constant $Z = p(y^*)$ explicitly, they can be viewed as implicitly solving the normalization problem, in the sense that their outputs behave as if such a normalization were available.

However, the presence of an explicit observable variable y is not essential. Rather, y serves as a convenient way to parameterize an unnormalized target distribution. From a more general perspective, posterior inference only requires access to an unnormalized probability (more precisely, a finite measure) $\tilde{p}(x)$. The objective is then to work with the corresponding normalized distribution

$$p(x) = \frac{1}{Z} \tilde{p}(x), \quad Z = \int_{\text{supp}(x)} \tilde{p}(x) dx, \quad (6)$$

either by explicitly estimating Z or, more commonly, by implicitly accounting for it through sampling or optimization procedures.

This formulation naturally accommodates cases in which $\tilde{p}(x)$ —or, more typically, $\log \tilde{p}(x)$ —is intractable and available only through unbiased stochastic estimators. A range of inference methods has been developed for such settings, including pseudo-marginal Metropolis–Hastings (Andrieu and Roberts, 2009), stochastic variational inference (Hoffman et al., 2013), and stochastic gradient Markov chain Monte Carlo (Nemeth and Fearnhead, 2020). We cast policy inference within this general framework.

Since we are interested in inferring a policy, the *policy* π is the latent random variable. The objective of MDP planning is to identify policies that maximize expected return. To align the probabilistic model with this objective, we assign to each policy an unnormalized **probability of optimality** that is monotone in its expected return.

Specifically, we define the unnormalized log probability of a policy as the *expected return* obtained by the agent following the policy in an episode, with expectation over trajectories τ_π distributed according to the dynamics induced by policy π :

$$\log \tilde{p}(\pi) = \mathbb{E}_{\tau_\pi} \sum_{t=1}^{H-1} R\left(s_t, a_t^{(\pi)}, s_{t+1}\right), \quad (7)$$

where $a_t^{(\pi)} = \pi(s_t)$ and $s_{t+1} = T\left(s_t, a_t^{(\pi)}\right)$ for all $t \in 1 \dots H$. This bears similarity to a Boltzmann–Gibbs distribution over trajectories, as in the maximum-entropy and control-as-inference literature (Ziebart et al., 2008; Todorov, 2006; Levine, 2018), however in our framework the policies rather than the trajectories are Boltzmann–Gibbs distributed.

Note that neither actions nor states are treated as Bayesian random variables for which a posterior is sought. Although the policy (if stochastic policies are considered) induces a stochastic action selection rule and the environment induces stochastic state transitions, these are generative rather than inferential sources of randomness: actions and state transitions are sampled forward from their respective distributions rather than conditioned for the purpose of posterior inference. The randomness they induce propagates into the estimation of the unnormalized log probability of the policy, so that $\log \tilde{p}(\pi)$ is available only through noisy Monte Carlo evaluations—by computing the return of a single episode:

$$\log \hat{\tilde{p}}(\pi) = \sum_{t=1}^{H-1} R\left(s_t, a_t^{(\pi)}, s_{t+1}\right). \quad (8)$$

Stochastic estimate (8) lays the basis for posterior inference of the policy distribution.

4 Inference

We begin by specifying the object of inference. In this work, we perform posterior inference over the space of **deterministic policies**, equipped with a

uniform prior ¹. While one may place a posterior over stochastic policies, doing so introduces an additional and largely redundant level of randomness. A posterior over stochastic policies combines epistemic uncertainty over policies with intrinsic action-level stochasticity, leading to mixtures of mixtures whose behavioral interpretation is unclear and whose posterior geometry is highly non-identifiable. In contrast, a posterior over deterministic policies represents uncertainty solely at the level of coherent behaviors, admits a clear decision semantics, and ensures that posterior concentration corresponds directly to behavioral certainty. Action stochasticity then arises naturally through marginalization over deterministic policies (Section 4.2.3), rather than being embedded inside each policy.

4.1 Algorithm

A natural baseline for posterior inference over the policy distribution specified by (8) is structured variational approximation (Hoffman et al., 2013). However, single-trajectory variational inference tends to produce underdispersed approximations, collapse onto a small subset of trajectory modes, and requires problem-specific heuristics to remain stable under multimodal dynamics or sparse rewards. The sequential structure of (8) encourages application of variational SMC, which multi-particle objective incorporates information from diverse trajectory proposals, yielding a tighter and more robust approximation to the policy posterior. These advantages, however, require adaptation of VSMC to the policy inference setting, which is the subject of this section.

VSMC optimizes a stochastic lower bound on the log marginal likelihood, where the stochasticity arises from Monte Carlo sampling of particle trajectories from the variational proposal. In the policy inference setting considered here, the objective is additionally noisy because the unnormalized log density of a policy is intractable and can only be evaluated through an unbiased Monte Carlo estimator (8). This second source of randomness is independent of the variational proposal and therefore does not alter the variational interpretation of the VSMC objective. It can be treated as exogenous estimator noise on the target density, analogous to the pseudo-marginal setting (Andrieu and Roberts, 2009), and incorporated directly into the optimization without modification of the underlying algorithm.

We constrain our setting to countable state spaces and finite action spaces. We require a countable state space to be able to identify previously visited states and to count the number of visits. We require a finite action space to be able to identify previously occurred state-action transitions and to count the number of occurrences, as well as to express policies by means of discrete distributions. In particular, for a deterministic policy space, a policy π in state s is an action

¹The proposed formulation also admits posterior inference over stochastic policies. We restrict attention to deterministic policies in this work to avoid an additional layer of action-level randomness and to keep the focus on policy-level uncertainty.

assignment, drawn from a categorical distribution over actions.

$$\pi(s) \equiv a|s \sim \text{Categorical}(\mathbf{p}(s)) \quad (9)$$

In a factored state space, where each state is represented by a fixed-dimensional vector of state variables, the distribution of policies can be parameterized by a neural network with the vector representation of the state as the input and the vector of action probabilities $\mathbf{p}$ as the output. We adopt this parametrization in our case studies.

SMC sweep Two adjustments to vanilla SMC sweep are required:

1. In order for particle policies to be deterministic, an action must be drawn from the proposal distribution on the first visit to each visited state only. If a trajectory revisits a state, both the prior and the proposal probability of the action assigned on the first visit are 1 (and of any other action are 0).
2. Because the stochastic estimate $\log \hat{Z}$ is induced by the entire particle system rather than by individual particles, stochasticity arising from environment dynamics must be coupled across particles. In particular, random state transitions must be shared, otherwise relative particle weights would be confounded by both differences in their policies and independent realizations of the transition noise. Concretely, whenever two particles visit the same state s and select the same action a for the same k th time, they are required to transition to the same successor state s' . This can be implemented by lazily sampling a shared transition tensor $\hat{T}$, where $\hat{T}_{s,a}^k \sim T(\cdot | s, a)$ is sampled once and reused by all particles within a sweep. The environment dynamics T are then replaced, during inference, by this shared random realization $\hat{T}$ ².

Optimization objective The original VSMC formulation assumes reparameterizable SMC proposal distributions and omits score function terms associated with the non-differentiable *resampling* operation due to their high variance and low signal. In finite-action MDPs, however, a categorical distribution over actions cannot be reparameterized. Consequently, while the resampling-induced score terms may still be dropped, the score-function contribution arising from *sampling actions from the proposal* must be retained in order to obtain meaningful gradients. This yields the surrogate objective

$$\mathcal{L} = \log \hat{Z} + \overline{\log \hat{Z}} \cdot \sum_{t=1}^H \sum_{i=1}^N \log q(a_{t,i} | s_{t,i}) \quad (10)$$

²This coupling of transition randomness is related to the classical **common random numbers** variance reduction technique (Mohamed et al., 2020), however here it is applied for correctness rather than for variance reduction.

where the overline denotes a stop-gradient operation. The high variance of this REINFORCE-style term can be substantially reduced by exploiting the temporal dependency structure of trajectories (Schulman et al., 2015). Defining

$$\log \hat{Z}_t = \sum_{t'=t}^H \log \left(\frac{1}{N} \sum_{i=1}^N w_{t',i} \right), \quad (11)$$

an equivalent, temporally stratified objective is obtained:

$$\mathcal{L} = \log \hat{Z} + \sum_{t=1}^H \left(\overline{\log \hat{Z}_t} \cdot \sum_{i=1}^N \log q(a_{t,i} | s_{t,i}) \right), \quad (12)$$

which replaces a single global learning signal with step-specific signals, thus improving estimator stability.

Algorithm 1 computes the surrogate VSMC objective for inference over deterministic policies. Certain performance-related refinements are omitted, for the sake of clarity and conciseness, from the algorithm pseudocode, and the reader is encouraged to consult the accompanying reference implementation for full detail.

4.2 Policy Selection

Variational policy inference yields an approximate posterior distribution over policies, represented by the proposal $q(a | s)$. To act in the environment, this posterior must be converted into a sequence of actions. This can be done in multiple ways, each corresponding to a different interpretation of policy uncertainty.

4.2.1 Deterministic Policy Extraction

A deterministic policy can be sampled *lazily* from the posterior by drawing an action the first time a state is encountered and subsequently reusing it. Algorithm 2 implements this procedure and induces a well-defined deterministic policy drawn from the posterior distribution.

In classical MDP planning, uncertainty over policies is typically discarded by selecting a single optimal policy—namely, the policy that maximizes expected return. From a Bayesian perspective, this corresponds to selecting a maximum *a posteriori* (MAP) policy under the inferred posterior.

In practice, MAP estimation is often approximated by annealing techniques that encourage posterior concentration around a dominant mode.

4.2.2 Annealing-Based MAP Selection

The variational objective underlying policy inference combines expected return with a regularization term measuring the discrepancy between the posterior and the prior. Increasing the relative weight of the return term therefore leads to a more concentrated posterior and, in the limit, to MAP-like behavior.

Algorithm 1 VSMC over deterministic policies

```

1: procedure DETERMINISTICPOLICYVSMC( $s_1$ , STEP, REWARD,  $H$ ,  $N$ ,  $q$ )
2:    $\log q_p \leftarrow 0$  ▷ proposal log probability
3:    $\log \hat{Z} \leftarrow 0$  ▷ log evidence estimate
4:    $M_A \leftarrow \emptyset$  ▷ action memory
5:    $M_C \leftarrow \emptyset$  ▷ transition count memory
6:    $M_T \leftarrow \emptyset$  ▷ transition memory
7:   for  $i \leftarrow 1, N$  do ▷ Initialize particles to  $s_1$ 
8:      $s^{(i)} \leftarrow s_1$ 
9:   end for
10:  for  $t \leftarrow 1, H$  do
11:    for  $i \leftarrow 1, N$  do
12:       $s \leftarrow s^{(i)}$ 
13:      ▷ Select action
14:      if  $(i, s) \in \text{dom } M_A$  then ▷ particle  $i$  previously visited  $s$ 
15:         $a \leftarrow M_A(i, s)$ 
16:         $c \leftarrow M_C(i, s, a)$ 
17:         $\log p_a, \log q_a \leftarrow 0, 0$ 
18:      else
19:         $a \sim q(s)$ 
20:         $M_A(i, s) \leftarrow a$ 
21:         $c \leftarrow 1$ 
22:         $\log p_a, \log q_a \leftarrow \frac{1}{|A|}, \log q(a \mid s)$ 
23:         $\log q_p \leftarrow \log q_p + \log q_a$ 
24:      end if
25:       $M_C(i, s, a) \leftarrow c + 1$ 
26:      ▷ Perform action
27:      if  $(s, a, c) \in \text{dom } M_T$  then ▷ any particle took  $a$  in  $s$   $c$  times
28:         $s' \leftarrow M_T(s, a, c)$ 
29:      else
30:         $s' \leftarrow \text{STEP}(s, a)$ 
31:         $M_T(s, a, c) \leftarrow s'$ 
32:      end if
33:       $w^{(i)} \leftarrow \text{REWARD}(s, a, s') + \log p_a - \log q_a$  ▷ Particle weight
34:    end for
35:     $\log \hat{Z} \leftarrow \log \hat{Z} + \log \sum_{i=1}^N \exp(w^{(i)})$ 
36:     $s^{(1:N)} \leftarrow \text{RESAMPLE}(s^{(1:N)}, w^{(1:N)})$ 
37:  end for
38:  return  $\log \hat{Z} + \overline{\log \hat{Z}} \cdot \log q_p$ 
39: end procedure

```

Algorithm 2 Drawing a deterministic policy from the posterior

```
1: procedure DRAWDETERMINISTICPOLICY( $s_1$ , STEP,  $H$ ,  $q$ )
2:    $M_A \leftarrow \emptyset$  ▷ action memory
3:    $s \leftarrow s_1$ 
4:   for  $t \leftarrow 1, H$  do
5:     if  $s \in \text{dom } M_A$  then ▷ agent previously visited  $s$ 
6:        $a \leftarrow M_A(s)$ 
7:     else
8:        $a \sim q(s)$ 
9:        $M_A(s) \leftarrow a$ 
10:    end if
11:     $s \leftarrow \text{STEP}(s, a)$ 
12:  end for
13: end procedure
```

One common approach is to anneal the reward scale during optimization. However, this strategy is undesirable for two reasons:

- Rewards are part of the domain specification and encode the agent’s preferences; modifying their scale alters the semantics of the problem.
- Increasing reward scale requires compensatory changes to optimization hyperparameters, introducing an artificial coupling between modeling assumptions and optimization dynamics.

An alternative is to anneal the regularization term by downscaling the prior-posterior discrepancy. With this modification, line 34 of Algorithm 1 becomes

$$w^{(i)} \leftarrow \text{REWARD}(s, a, s') + T \cdot (\log p_a - \log q_a), \quad (13)$$

where the temperature T decreases from 1 to 0. This procedure is formally equivalent to increasing the relative weight of rewards and admits an interpretation as progressively collapsing the prior onto the inferred posterior, reminiscent of variational policy search methods (van de Meent et al., 2016).

While effective for identifying a single high-probability policy, annealing ultimately discards information encoded in the absolute scale of rewards.

4.2.3 Stochastic Policy Selection via Posterior Predictive Sampling

Rather than collapsing the posterior, we may instead interpret it as a representation of uncertainty over policies. In this view, rewards encode the agent’s preferences, and differences in reward magnitude reflect the degree of certainty associated with those preferences.

Under this interpretation, each policy drawn from the posterior may be viewed as optimal for a particular realization of the agent’s preferences, with posterior mass reflecting plausibility. Action selection can then be performed

by sampling from the posterior predictive distribution, corresponding to recurrent Thompson sampling: at each decision point, a policy is sampled from the posterior and an action is drawn from that policy.

For a finite action space, this is equivalent to sampling an action directly from $q(s)$ at each time step, as shown in Algorithm 3.

Algorithm 3 Drawing a stochastic mixed policy from the posterior

```

1: procedure DRAWSTOCHASTICPOLICY( $s_1$ , STEP,  $H$ ,  $q$ )
2:    $s \leftarrow s_1$ 
3:   for  $t \leftarrow 1, H$  do
4:      $a \sim q(s)$ 
5:      $s \leftarrow \text{STEP}(s, a)$ 
6:   end for
7: end procedure

```

As reward differences increase, the posterior concentrates, and the resulting mixed policy approaches a deterministic one. Conversely, when rewards are comparable, the mixed policy remains diffuse, reflecting genuine uncertainty in action selection. In this sense, the resulting behavior is **optimal under preference uncertainty**, retaining information that is lost under annealing-based MAP selection.

5 Related Work

This work connects to two broad and partially overlapping lines of research:

- (i) formulations of control and planning as probabilistic inference, and
- (ii) entropy-regularized reinforcement learning and policy search methods.

While both perspectives are well established, the present work differs from prior approaches in the *probabilistic object being inferred*—a posterior over policies—and in the resulting implications for stochasticity, entropy regularization, and inference algorithms.

5.1 Control and Planning as Inference

The idea of casting decision making and control as probabilistic inference has a long history. Early formulations include Attias (2003), who introduced planning as inference in graphical models, and later developments by Botvinick and Toussaint (2012); Toussaint and Storkey (2006), and related work that formalized optimal control problems as inference in structured probabilistic models. In these approaches, optimality is typically encoded via auxiliary variables or likelihood terms that bias trajectories toward high return.

A closely related and influential line of work is the *active inference* framework, which models action selection as approximate Bayesian inference under

a generative model that includes preferences over outcomes. While differing in interpretation and objectives, these methods similarly introduce probabilistic structure to express control objectives.

More recently, *control-as-inference* formulations popularized in the reinforcement learning community—most notably in the work of Levine and colleagues—have shown that entropy-regularized RL objectives can be derived as variational inference problems (e.g., Levine (2018)). These formulations typically introduce either fictitious observations or optimality variables, such that maximizing expected return corresponds to maximizing the likelihood of optimality under a suitably defined model.

The present work adopts the control-as-inference viewpoint but differs in a fundamental modeling choice. Rather than introducing auxiliary observations or trajectory-level optimality variables, we treat the *policy itself* as the latent random variable and interpret the expected return of a policy as its unnormalized log probability. This yields a posterior distribution over policies directly, without augmenting the model with additional observation channels. As a result, the Boltzmann distribution over policies arises naturally from the probabilistic interpretation, rather than being imposed through likelihood design or entropy penalties.

5.2 Entropy-Regularized Reinforcement Learning and Policy Search

A second closely related body of work concerns entropy-regularized reinforcement learning and stochastic policy optimization, including policy gradient methods (Sutton et al., 1999), soft Q-learning (Haarnoja et al., 2017), and Soft Actor-Critic (SAC) (Haarnoja et al., 2018; Christodoulou, 2019). These methods optimize objectives of the form

$$\mathbb{E}[R(\pi)] + \alpha \mathcal{H}(\pi), \quad (14)$$

where the entropy term encourages exploration and stabilizes learning.

Several works have established equivalences between these approaches. In particular, Schulman et al. (2017) showed that policy gradients and soft Q-learning optimize the same underlying objective under appropriate parameterizations. From the control-as-inference perspective, entropy regularization is often interpreted as arising from a variational bound, with the temperature parameter controlling the trade-off between return and entropy.

While mathematically related, the interpretation in the present work differs in three key respects. First, entropy regularization is not introduced as a heuristic or tuning mechanism, but as a direct consequence of placing a prior over policies. In particular, a uniform prior leads to entropy-regularized objectives, while alternative priors yield different regularization structures.

Second, stochasticity in prior approaches is typically a property of the learned policy itself. In contrast, stochasticity here reflects *posterior uncertainty over deterministic policies*. Actions are sampled by first sampling a policy from the

posterior and then executing it, rather than by directly sampling actions from a stochastic policy parameterization. This distinction aligns the method with Thompson sampling, though in a structured, state-conditional form appropriate for sequential decision making.

Third, the inference formulation enables the use of **variational sequential Monte Carlo (VSMC)** (Naesseth et al., 2018) as a natural inference algorithm. Unlike single-trajectory variational inference commonly used in RL, VSMC maintains and resamples a population of policy hypotheses, allowing posterior mass to concentrate on high-reward regions of policy space while preserving uncertainty. As noted in prior work on variational SMC, several existing methods emerge as limiting cases: single-particle VSMC reduces to structured variational inference, while multi-particle methods without resampling correspond to importance-weighted objectives. Within the present framework, previously introduced entropy-regularized algorithms can be recovered as special or degenerate cases of the proposed inference procedure.

Taken together, these connections position the proposed method not just as an alternative optimizer for existing RL objectives, but as a unifying probabilistic framework in which entropy regularization, exploration, and policy uncertainty arise from explicit Bayesian modeling choices.

6 Experiments

6.1 Grid Worlds

6.2 Relational Factored Domains

6.2.1 Academic Advising

6.2.2 Elevators

6.2.3 Wildfire

7 Discussion

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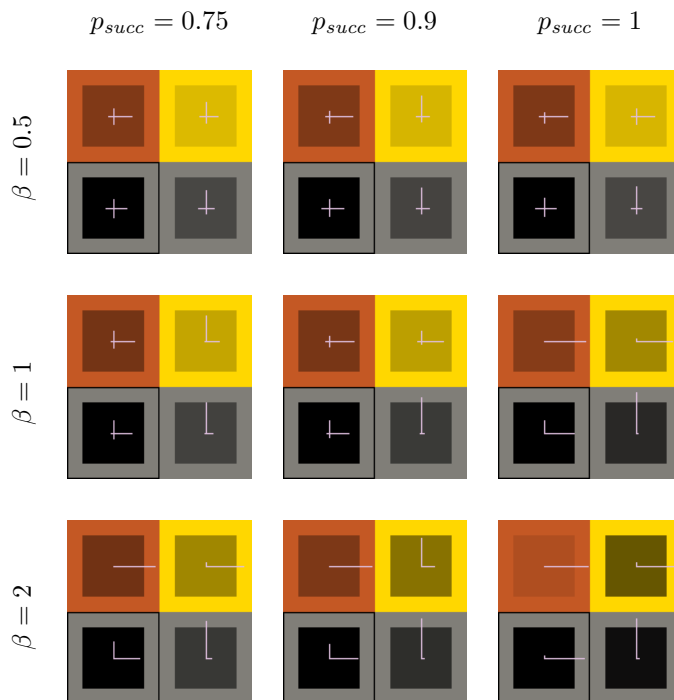


Figure 1: Grid worlds: influence of reward scale β and action success probability p_{succ} on the policy distribution. The higher the reward scale, the more concentrated the policy distribution is around the mode. Action success probability affects both the occupancy and the policy distribution.

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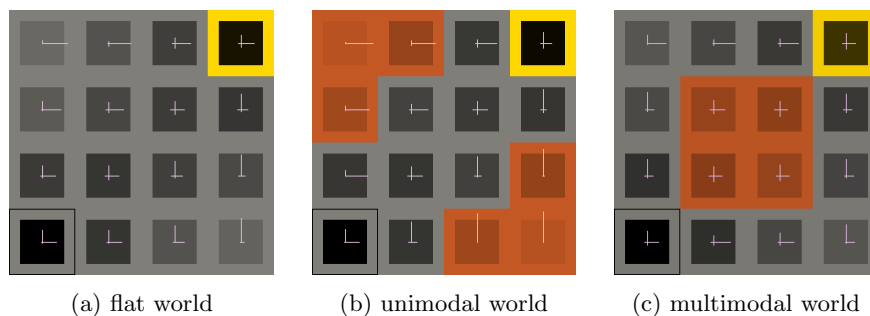


Figure 2: Grid worlds: policies and occupancies.

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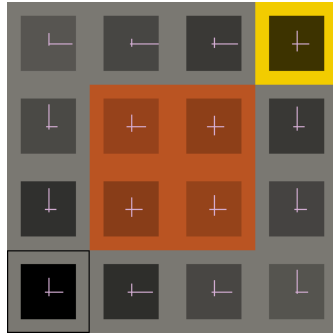
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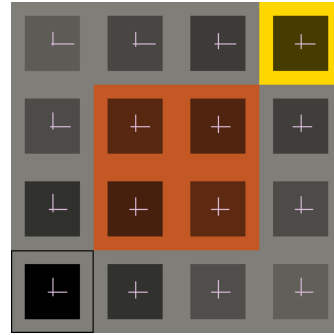
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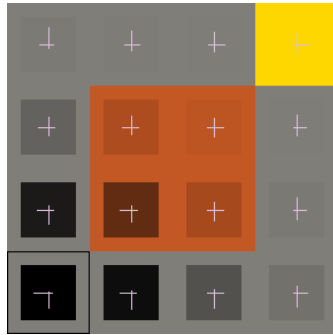
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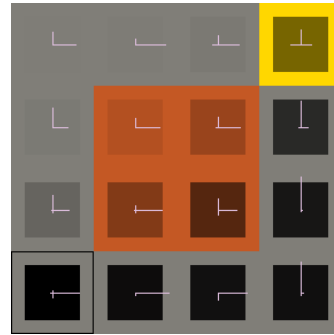
(a) variational SMC



(b) variational importance sampling



(c) variational structured approximation, common outcome



(d) variational structured approximation, rare outcome

Figure 3: Grid worlds: VSMC variations on multimodal world.

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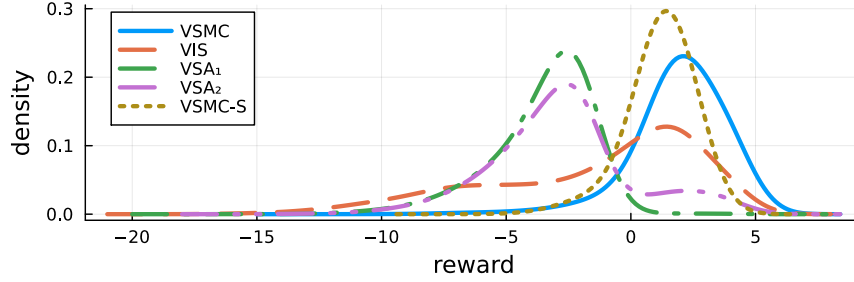


Figure 4: Trajectory return distribution of VSMC variants.

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